॥ श्रीगणेशाय नमः ॥

Ratios, Proportions, and Partnerships

$a : b$ ← consequent
↑ antecedent

Proportion:
equality of ratios. $a : b :: c : d \Rightarrow \frac{a}{b} = \frac{c}{d}$

$a : b : c = A : B : C \Leftrightarrow \frac{a}{A} = \frac{b}{B} = \frac{c}{C}$

Types of ratios

1. Duplicate ratios:
a:b = p:q. Duplicate ratio of a:b is given by $p^2:q^2$.

2. Triplicate ratios:
a:b = p:q. Triplicate ratio of a:b is $p^3:q^3$.

3. Sub-duplicate ratios:
a:b = p:q. Sub-duplicate ratio of a:b is given by $\sqrt{p}:\sqrt{q}$.

4. Sub-triplicate ratios:
a:b = p:q. Sub-triplicate ratio of a:b is given by $\sqrt[3]{p}:\sqrt[3]{q}$.

5. Inverse ratios:
a:b = p:q. Inverse ratio of a:b is given by q:p.

6. Compound ratios:

$a : b = 1 : 3$
$c : d = 2 : 9$
$e : f = 4 : 7$

compound ratio $= \frac{1 \cdot 2 \cdot 4}{3 \cdot 9 \cdot 7} = 8 : 189$

NOTE: Whenever ratios are in fractions, convert them to integers. $\frac{2}{9} : \frac{1}{3} \rightarrow \frac{2}{9} \times 9 : \frac{1}{3} : 9 \rightarrow 2 : 3$

Q. If a:b:c = 2:3:4, then what is the value of a/b:b/c:c/a?

→

$\frac{a}{b} : \frac{b}{c} : \frac{c}{a} = \frac{2}{3} : \frac{3}{4} : \frac{4}{2}$
$8 : 9 : 24$

Q. If a:b=2:3, b:c=4:5,

$2 : 3 = 8 : 12$
$4 : 5 = 12 : 15$

$a : b : c = 8 : 12 : 15$

a	b	c
2	3 →	3
4 ←	4	5
8	12	15

Q. $a : b = \frac{1}{2} : \frac{1}{3}$ $b : c = \frac{1}{2} : \frac{1}{3}$

a	b	c
$\frac{1}{2}$	$\frac{1}{3}$	$\frac{1}{3}$
$\frac{1}{2}$	$\frac{1}{2}$	$\frac{1}{3}$
$\frac{1}{4}$	$\frac{1}{6}$	$\frac{1}{9}$

$9 : 6 : 4$

Q. If $a:b = 5:7$, $c:d = 2a:3b$. Then $ac:bd = ?$

→ $\frac{ac}{bd} = \frac{5}{7} \cdot \frac{2a}{3b} = \frac{10a}{21b} = \frac{10}{21} \cdot \frac{5}{7} = \frac{50}{147}$

Q. $3a = 5b$, $4b = 6c$, then $a:c = ?$

→ $\frac{a}{b} = \frac{5}{3}$ $\frac{b}{c} = \frac{3}{2}$ $a:b:c = 5:3:2$

$a:c = 5:2$

Q. $a:5 = b:7 = c:8$. $\frac{a+b+c}{a} = ?$

→ $\frac{a}{5} = \frac{b}{7} = \frac{c}{8}$.

$\frac{a+b+c}{a} = 1 + \frac{b}{a} + \frac{c}{a} = 1 + \frac{7}{5} + \frac{8}{5} = \frac{5+7+8}{5} = 4$

Q. $\frac{x}{y} = \frac{6}{5}$. $\frac{x^2+y^2}{x^2-y^2} = ?$

→ $\frac{36k^2 + 25k^2}{36k^2 - 25k^2} = \frac{61}{11}$

Q. $\frac{x}{24} = \frac{6}{7}$. $\frac{x-y}{x+y} + \frac{14}{19} = ?$

→ $\frac{x}{y} = \frac{12}{7}$

$\frac{x+y}{x-y} = \frac{12-7}{12+7} = \frac{5}{19}$ $\frac{5}{19} + \frac{14}{19} = 1$

Q. $a:b:c = 8:12:15$. $a^2:b^2:bc = ?$

→ $64:144:180$

$= 16:36:45$

Q. If ₹1000 are divided into 2 people in the ratio 3:2, what will each get?

→ $\frac{3}{5} \cdot 1000$ and $\frac{2}{5} \cdot 1000$

=

Q. If 78 is divided into 3 parts in the ratio 1:1/3:1/6, what's the 2nd part?

→ $1 : \frac{1}{3} : \frac{1}{6}$

$6 : 2 : 1$

$\frac{2}{9} \times \frac{26}{78} = \frac{52}{3}$

Q. If ₹33630 are divided among a, b, and c s.t. a:b=3:7, b:c=6:5, what's the amount received by b?

→

a	b	c
3	7	7
6	6	5

$18:42:35$

$\frac{42}{95} \times 33630 = ₹14868$

Q. ₹3400 are divided among a, b, c, and d s.t. a:b=2:3, b:c=4:3, c:d=2:3, what's the sum of shares of b and d?

→

2	3	3	3
4	4	3	3
2	2	2	3

$16:24:18:27$

$\frac{24}{85}(3400) + \frac{27}{85}(3400) = \frac{51}{85} \cdot 3400 = \frac{102000}{5} = 20400$

Q. By mistake, instead of dividing Rs. 117 among a, b, and c in the ratio 1/2:1/3:1/4, it was divided in the ratio 2:3:4. Who gains the most and by how much?

→ $\frac{1}{2} : \frac{1}{3} : \frac{1}{4}$

$6 : 4 : 3$

should have done this

$\frac{6}{13} \cdot 117, \frac{4}{13} \cdot 117, \frac{3}{13} \cdot 117$

$54, 36, 27$

$2 : 3 : 4$

Went actually did this

$\frac{2}{9} \cdot 117, \frac{3}{9} \cdot 117, \frac{4}{9} \cdot 117$

$26, 39, 52$

a gained $26 - 54 = -28$. b gained $39 - 36 = 3$

c gained $52 - 27 = 25$.

c gains most by 25.

Q. If a sum is to be divided among a, b, c s.t. a's share is twice of b's. b's is 4 times that of c. What's the ratio of their shares?

→ $a = 2b$

$\frac{a}{b} = \frac{2}{1}$ $\frac{b}{c} = \frac{4}{1}$

2	1	1
4	4	1
8	4	1

Q. Divide Rs. 1250 among a, b, c s.t. a gets 2/9 of b's share and c gets 3/4 of a's share.

→ $a = \frac{2}{9} b$ $c = \frac{3}{4} a$

$a : b = 2 : 9$ $c : a = 3 : 4$

c	a	b
3	4	
	2	1
6	8	36
3	4	18

$a : b : c = 4 : 18 : 3$

$\frac{4}{25} \cdot 1250, \frac{18}{25} \cdot 1250, \frac{3}{25} \cdot 1250$

$4 \cdot 50, 18 \cdot 50, 3 \cdot 50$

$200, 900, 150$

Q. Rs.2010 are to be divided among a, b, c s.t. that if a gets rs.5 then b must get rs.12 and if b gets rs.4 then c must get rs.5.5. Share of c exceeds that of b by ___.

→ $\frac{a}{b} : \frac{5}{12}$ $\frac{b}{c} = \frac{4}{5.5} = \frac{8}{11}$

5	12	
	8	11
40	96	132
10	24	33

$C - B = \left(\frac{33 - 24}{67}\right) \cdot 2010 = 9 \cdot 30 = \underline{\underline{270}}$

Q. The ratio of 2 #s is 3:8 and their diff is 115. The larger # is ___.

→ $8x - 3x = 115$

$5x = 115$

$x = 23$ $8x = \underline{\underline{184}}$

Q. The ratio of 2 #s is 10:7 and their diff is 105. Their sum is ___.

→ $3x = 105$

$x = 35$ $17x = 17 \cdot 35 =$

Q. The ratio of 3 #s is 3:2:5 and the sum of their sqs is 1862. The #s are?

→ $9x^2 + 4x^2 + 25x^2 = 1862$

$x^2 = 49$

$x = 7$

$3x, 2x, 5x$

$21, 14, 35$

Q. a/b = b/c = c/d = 3/4. If b gets rs. 308 less than d, then their shares are?

→ $\frac{a}{b} = \frac{b}{c} = \frac{c}{d} = \frac{3}{4}$.

a : b
b : c
c : d

3 4 4 4
3 3 4 4
3 3 3 4
27 : 36 : 48 : 64

$64x - 36x = 308$

$x = 11$

$\frac{308}{11} = 28$

shares : $27:11$, $36:11$, $48:11$, $64:11$

297, 396, 528, 704

Q. The total # of students of a school was 660. The ratio of boys to girls is 13:9. After some days 30 girls joined and some boys left. The ratio becomes 6:5. What's the # of boys that left?

→ $\frac{b}{g} = \frac{13}{9}$

$b + g = 660$
$13x + 9x = 660$
$22x = 660$
$x = 30$

$b = 390$
$g = 270$

$\frac{390 - y}{300} = \frac{6}{5}$

$390 - y = 360$
$y = 30$

Q. Rs.2600 is to be divided among a, b, c s.t. 2a=3b=4c. What's the share of a?

→ $2a = 3b = 4c$

$\frac{a}{1/2} = \frac{b}{1/3} = \frac{c}{1/4}$ $\quad a : b : c = 6 : 4 : 3$

$\frac{6}{13} \cdot 2600 = 1200$

Q. Students in 3 classes are in the ratio 2:3:5. If 40 students are increased in each, the ratio changes to 4:5:7. Originally, the total # of students was ___.

→ $2k \quad 3k \quad 5k$

$2k + 40 : 3k + 40 : 5k + 40 = 4 : 5 : 7$

$\frac{2k + 40}{3k + 40} = \frac{4}{5} \Rightarrow 10k + 200 = 12k + 160$

$40 = 2k$

$k = 20$

total # of students $= 20 \cdot 10 = 200$

Q. The ratio of # of boys to that of girls in a group becomes 2:1 when 15 girls leave. Afterwards, when 45 boys leave, it becomes 1:5. Find the initial # of girls in the group.

→ $\frac{b}{g - 15} = \frac{2}{1} \Rightarrow b = 2g - 30$

$\frac{b - 45}{g - 15} = \frac{1}{5} \Rightarrow (2g - 30 - 45)5 = g - 15$

$9g = 360$

$g = 40$

Q. There are 48 students. B:G=5:3. Some girls came. B:G=6:5. How many girls came?

→ $b = 30 \quad g = 18$

$\frac{30}{18 + x} = \frac{6}{5} \Rightarrow x = 7$

Incomes, savings, and expenditures

Income = expenditure + savings

Q. Ratio of incomes of P and Q is 3:4. The ratio of their expenditures is 2:3. If both of them save Rs.6000 each, what's the income of P?

→ Incomes : $3i, 4i$

$$\frac{3i-6000}{4i-6000} = \frac{2}{3}$$

$$9i - 18{,}000 = 8i - 12{,}000$$

$$i = 6{,}000$$

$$3i = \underline{\underline{18{,}000}}$$

Q. Incomes, A:B=5:6. Exp, 3:4. If savings are 1800 and 1600 resp. find inc(B).

→ $5i, 6i$

$$\frac{5i-18}{6i-16} = \frac{3}{4} \Rightarrow 20i - 72 = 18i - 48$$

$$2i = 72 - 48$$

$$i = 12 \hookrightarrow 1200$$

$$6i = 7200$$

Q. Exp:savings = 26:3. Monthly income is 7250. Monthly savings = ?

→ Exp = $26x$, Sav = $3x$

$$7250 = (26+3)x$$

$$x = 250$$

$$3x = \underline{\underline{750}}$$

Q. Inc(A):Inc(B) = 5:3. Exp(A):Exp(B):Exp(C)=8:5:2. C spends rs.2000 and B saves rs.700, then A saves ____.

→ Exp(C) = 2000

Exps : 8 : 5 : 2

$$2x = 2000$$

$$x = 1000$$

Exps : 8000, 5000, 2000

B saves 700

Say A saves y.

Inc = exp + sav.

$$\frac{5}{3} = \frac{8000+y}{5000+700} \Rightarrow \frac{5(5700) - 24000}{3} = y$$

$$y = \underline{\underline{1500}}$$

Q. Sal of A, B, C = 2:3:5. C's sal is 12,000 more than A's. B's sal is ___.

→ $2x$ $3x$ $5x$

$$5x = 2x + 12{,}000$$

$$x = 4{,}000$$

$$3x = 12{,}000$$

Coins

Q. Ratio of incomes of P and Q is 3:4. The ratio of their expenditures is 2:3. If both of them save Rs.6000 each, what's the income of P?

Partnerships

If the partners' investment ratio is p1:p2:p3 and the ratio of investment times is t1:t2:t3, then the ratio of profits is given by p1t1:p2t2:p3t3.

Q. If A, B, C invest the same amount over 3 years, 2 years, and 5 years resp, what should be the ratio of their profits?

→ 3:2:5

If time is same → ratio of profit = ratio of investment.
If amount is same → ratio of profit = ratio of time.

Q. Ajay started a business investing Rs.12,000. After 5 months, Bala joined him with a capital of Rs.9000. At the end of the year, the total profit is Rs.4416. What is the share of Ajay?

→ A 5 months, B 7 months

Ratio of profits = $12000 \times 12 : 9000 \times 7$

$= 12 \times 12 : 9 \times 7$

$= 4 \times 4 : 7$

$= 16 : 7$

Ajay's share = $\frac{16}{23} \times 4416 = 3072$

Q. Arun started a business investing Rs.45,000. After 8 months, Vani joined in with Rs.52,000. At the end of 2 years, the profit was Rs.56,165. What is Vani's share?

→ $24 - 8 = 16$

$45 \times 3 : 16 \times 2$

$135 : 104$

$\frac{104}{239} \times 56165 = 24440$

Q. A started with Rs.45,000. B joined later with Rs.54,000. If profit at the end of the year was divided in the ratio 2:1, when did B join?

→ $45 \cdot 12 : 54 \cdot x = 2 : 1$

$10 : x = 2 : 1$

$x = 5$

B's time = 5 months.

∴ B joined after 7 months.

Q. Ram started with Rs.10k. Banu joined later with Rs.40k. If both get an equal share at the end, when did Banu join?

→ $10 \times 12 : 40 \times x = 1 : 1$

$x = 3$

9 months.

Q. Ajay, Adhi, and Arun invested Rs.45k, 70k, and 90k resp. to start a business. At the end of 2 years, profit is Rs.1.64 lakhs. What is Adi's share?

→ $\frac{70}{205} \times 164 = 56$

₹56000

Q. A started a business investing Rs.45k. After 3 months, B joined with Rs.60k. After another 6 months, C joined with Rs.90k. At the end of the year, the profit is Rs.16,500. Find their shares.

→ A = 12 months
B = 9
C = 3

$45 \times 12 : 60 \times 9 : 90 \times 3$

$2 : 2 : 1$

$\frac{2}{5} \times 16500 = 6600, \quad \frac{1}{5} \cdot 16500, \quad \frac{1}{5} \cdot 16500$

6600, 6600, 3300

Q. Arun started a business with Rs.32k. After 5 months, Bala joined with 22k. Profit at eoy is Rs.16409. Diff betn shares?

→ $32 \cdot \frac{12}{6} : 22 \cdot 7$

$192 : 77$

$\frac{192-77}{192+77} = \frac{115}{269}$

$\frac{115}{269} \cdot 16409 = 7015$

Q. A, B, C enter into a partnership. They invest Rs.40k, 80k, 120k resp. At eoy, B withdraws Rs.40k. At the end of 2nd year, C withdraws Rs.80k. In what ratio will the profit be shared at the end of 3 years?

→ A : 40K for 3 yrs.

B : 80K for 1yr, 40K for 2yrs

C : 120K for 2yrs, 40K for 1yr.

$(40 \times 3) : (80 + 40 \times 2) : (120 \times 2 + 40 \times 1)$

$120 : 160 : 280$

$3 : 4 : 7$

Q. A and B enter into a partnership for a year. A contributes Rs.1500 and B contributes Rs.2000. After 4 months, they admit C, who contributes Rs.2250. If B withdraws his contribution after 9 months, how would the profits (Rs.900) be shared at eoy?

→ A → 1500 for 12 months

B → 2000 for 9 months

C → 2250 for 8 months.

$\frac{15}{6} \times \frac{12}{3} : \frac{20}{8} \times 9 : \frac{225}{9} \times \frac{8}{2}$

$18 : 18 : 18$

$1 : 1 : 1$

$300 : 300 : 300$

Q. A, B, C invested Rs.42k, 30k, 28k. After 4 months, A withdrew 12k, B 6k, C 8k. After 10 months, profit = 46420. Share of C?

→ A → 42k, 30k

B → 30, 24

C → 28, 20

4 months, 6 months

$42 \times 4 + 30 \times 6 : 30 \times 4 + 24 \times 6 : 28 \cdot 4 + 20 \cdot 6$

$348 : 264 : 232$

$\frac{232}{844} \times 96420 = 12760$

Q. A, B, C divide Rs.4200 among themselves in the ratio 7:8:6. If Rs.200 is added to each of their shares, what'll be the new ratio?

→ $\frac{7}{21} \cdot 4200 + 200 : \frac{8}{21} \cdot 4200 + 200 : \frac{6}{21} \cdot 4200 + 200$

$1400 + 200 : 1600 + 200 : 1200 + 200$

$16 : 18 : 14$

$8 : 9 : 7$

Q. P, Q, R start a business. Twice the capital of P = Thrice the capital of Q. Q = 4R. Find the share of Q in an annual profit of Rs.1,48,500.

→ $2P = 3Q$ $\quad Q = 4R$

$\frac{P}{Q} = \frac{3}{2} = \frac{6}{4}$ $\quad \frac{Q}{R} = \frac{4}{1}$

$6 : 4 : 1$

$\frac{4}{11} \times 148500 = 13500 \cdot 4 = 54000$

Q. A invests some amount at the beginning. B invests double the amount of A after 6 months. C invests 3 times A after 8 months. If the annual profit is 54,000, what's C's share?

→ A → $x \times 12$

B → $2x \times 6$

C → $3x \times 4$

$1 : 1 : 1$

$\frac{54000}{3} = 18000$

Q. A and B start a business by investing in the ratio 4:3. If 9% of the total profit goes to charity, and A's share is 1196, find total profit.

→ $\frac{4}{7}\left(\frac{91}{100} x\right) = 1196$

$x = \frac{1196 \times 700}{91 \times 4} = 2300$

Q. A and B are partners. A contributes 1/4th of the capital for 15 months. B received 2/3rd of the profits. How long was B's money used?

→ Investment : 1 : 3
for time : 15 : x
profit : 15 : 3x

$\frac{3x}{15+3x} = \frac{2}{3}$

$9x = 30 + 6x$

$x = 10$

10 months.

Q. A and B started a business with initial investments in the ratio 12:11. Annual profit 4:1. If A invested for 11 months, B?

→ $12 \times 11 : 11 \times x = 4 : 1$

$12 : x = 4 : 1$

3 months.

Q. A and B started a business with initial investments 20k and 35k. They agreed to share the profit in the ratio of capital at eoy. C joins on the condition that A, B will share profits equally. C pays 220k as premium for this to be shared betn A and B. This is to be divided betn A and B in the ratio of ___.

→

A	B
(20K×12) + (x×12)	(35K×12) + (220−x)12

$20 \cdot 12 + 12x = 35 \cdot 12 + 220 \cdot 12 - 12x$

$24x = 35 \cdot 12 + 220 \cdot 12 - 20 \cdot 12$

$x = 117.5$

$220 - x = 102.5$

$1175 : 1025$

$47 : 41$ ×

(20×12) → 240

(35×12) → 420

total at the end is to be divided equally betn A and B:

240
420
220
880

$\frac{880}{2} = 440$ for A and B each.

So amount initially is 240 and 420. We need some more.

440 − 240 = 200

440 − 420 = 20

200 : 20

10 : 1

Practice session

Q. X varies directly as Y and X varies inversely as the square of 2. When Y=75 and X=6, the value of Z is 5. Find X if Y=24 and Z=4.

→ $X \propto \frac{Y}{Z^2}$ $\quad x = k\frac{Y}{Z^2}$ $\quad x = 2\cdot\frac{24}{16} = 3$

$6 = k\cdot\frac{75}{25}$

$k = 2$

Approach 2: $X \propto \frac{Y}{Z^2}$ $\quad \frac{x_1}{x_2} = \frac{\frac{y_1}{z_1^2}}{\frac{y_2}{z_2^2}}$

Q. x varies as y directly and as z inversely. x=14 when y=10. Find z when x=49, y=45.

→ $x \propto \frac{y}{z}$ $\quad \frac{2\cancel{14}}{7\cancel{49}} = \frac{10}{z_1}\cdot\frac{z_2}{45}$ $\quad \Rightarrow$ CANNOT BE DETERMINED.

$\frac{2}{70} = \frac{z_2}{z_1}$

Q. If $\frac{a}{b+c} = \frac{b}{a+c} = \frac{c}{a+b} = \frac{k}{9}$, find k.

→ If $\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} = \frac{m}{n} \Rightarrow \frac{a_1+a_2+a_3}{b_1+b_2+b_3} = \frac{m}{n}$

$\therefore \frac{a+b+c}{2(a+b+c)} = \frac{k}{9}$

$\Rightarrow k = 9/2$

Q. $2x+3y+4z = 27$, $2x:3y:4z = 1:2:3$. Find $x^1y^2z^3$

→ $4k+2k+3k = 9k = 27$

$k = 3$

$2x = 4k = 12$ $\quad 3y = 2k = 6$ $\quad 4z = 3k = 9$

$x = 6$ $\quad y = 2$ $\quad z = \frac{9}{4}$

$\frac{\cancel{3}\cdot\cancel{3}\cdot\cancel{3}\cdot\cancel{3}\cdot\cancel{2}\cdot\cancel{2}\cdot 9\cdot 9\cdot 9}{\cancel{4}\cdot\cancel{4}\cdot 4} = 3^{10}$

$a:b::b:c$ ↓ $\quad a:b::b:c$ ↑

Q. third proportional of 21 and 42 is x. mean proportional of 49 and 121 is y. $x:y = $ ____.

→ $21:42::42:x$

$1:2::42:x$

$x = 84$

$49:y::y:121$

$\frac{49}{y} = \frac{y}{121}$

$y^2 = 49\cdot 121$

$y = 7\cdot 11 = 77$

$x:y = 84:77$

$= 12:11$

Q. Find 4th proportional to 60, 48, 30.

→ $\frac{60}{48} = \frac{30}{x} \Rightarrow x = 24$

Q. The duplicate ratio of $\frac{x+3}{x+7}$ is $\frac{4}{9}$.

→ $\frac{(x+3)^2}{(x+7)^2} = \frac{4}{9}$ $\quad \frac{x+3}{x+7} = \frac{2}{3}$

$3x+9 = 2x+14$

$x = 5$

Q. Sub-duplicate ratio of $x-1$ and $x+3$ is 2:3.

→ $\frac{x-1}{x+3} = \frac{4}{9}$

$9x-9 = 4x+12$

$5x = 21$

$x = \frac{21}{5} = 4.2$

Q. A square and an equilateral Δ have equal side lengths. Ratio of area(sq) : area(Δ) = ?

→ $$\frac{a^2}{\frac{\sqrt{3}}{4}a^2} = \frac{4}{\sqrt{3}}$$

Q. If the perimeter of a sq. and an eq. Δ are equal, what is the ratio of their areas?

→ $4a_1 = 3a_2$

$$\frac{a_1}{a_2} = \frac{3}{4}$$

$$\frac{\text{area of sq}}{\text{area of eq }\Delta} = \frac{9}{\frac{\sqrt{3}}{4}16} = 9 : 4\sqrt{3}$$

Q. $\frac{10a^2+ab}{3ab-b^2} = \frac{10}{1}$. $a:b = ?$

→ $10a^2 + ab = 30ab - 10b^2$

Divide by ab,

$$10\frac{a}{b} + 1 = 30 - 10\frac{b}{a}$$

$$10x + 1 = 30 - \frac{10}{x}$$

$$10x + \frac{10}{x} = 29$$

$$10\left(x + \frac{1}{x}\right) = 29$$

$$\frac{x^2+1}{x} = \frac{29}{10}$$

$$10x^2 - 29x + 10 = 0$$

$$10x^2 - 25x - 4x + 10 = 0$$

$$5x(2x-5) - 2(2x-5) = 0$$

$$x = \frac{2}{5}, \frac{5}{2}$$

$$x = \frac{a}{b} \qquad a:b = 2:5 \text{ or } 5:2$$

Q. A and B start a business jointly. A invests 16K for 8 months. B remains in the business for 4 months. Of the total profit, B claims $\frac{2}{7}$ of the profit. How much money did B contribute?

→ A:B profits = 5:2

$$\frac{16 \times 8^2}{x \times 4} = \frac{5}{2}$$

$$x = \frac{64}{5} = 1.28 \qquad 12800$$

Q. A father wanted to divide some amount among A, B, C in ratio 3:4:5. By mistake, he divided it s.t. $3a = 4b = 5c$. Who among A, B, C were in profit due to the mistake?

→ $3a = 4b = 5c$

$$\frac{a}{\frac{1}{3}} = \frac{b}{\frac{1}{4}} = \frac{c}{\frac{1}{5}}$$

$$\frac{a}{20} = \frac{b}{15} = \frac{c}{12}$$

20 : 15 : 12 3 : 4 : 5

$\frac{20}{47}$ $\frac{31}{124}$ $\frac{15}{47}$ $\frac{41}{123}$ $\frac{12}{47}$ $\frac{5}{12}$

$>$ $<$ $\frac{144}{235}$ $<$

Q. For the same ques, if C got a diff. in amount of ₹546 due to this mistake, find the amount that was divided.

→ $\left|\frac{12}{47}x - \frac{5}{12}x\right| = 546$

$$\left|\frac{144 - 235}{47 \cdot 12}\right| \cdot x = 546$$

$$x = \frac{546 \cdot 47 \cdot 12}{91} = 3384$$

Q. A, B, C became partners in a business by investing money in the ratio 5:7:6. Next yr they increase the investments by 26%, 120%, 15% resp.

In what ratio should profits of 2nd yr be divided?

→ 5:7:6

$$5 + \frac{26}{100} \cdot 5 : 7 + \frac{120}{100} \cdot 7 : 6 + \frac{15}{100} \cdot 6 = 630 : 1540 : 690$$

Q. A, B, C enter into a partnership $\frac{7}{2} : \frac{4}{3} : \frac{6}{5}$. After 4 months, A increases share by 50%. If total profit at end of 1yr is 21600, then B's share is ___.

→ For 4 months : 105 : 40 : 36

For 8 months : 150% 105 : 40 : 36

Profits : $105 \times 4 + \frac{150}{100} \cdot 105 \times 8 : 40 \times 12 : 36 \times 12$

$105 \cdot 16 : 40 \cdot 12 : 36 \cdot 12$

$105 : 10 \cdot 3 : 9 \cdot 3$

$35 : 10 : 9$

B's share : $\frac{10}{54} \times 21600 = 4000$

Q. The ratio betn 2 #s is 3:4, their LCM is 180. The 1st # is ?

→ $3x \cdot 4x = 180 \times x$ (HCF)

$x = \frac{180}{12} = 15$

$x = 15$

1st # $= 3x = 45$.

Q. The value of a diamond varies directly with the cube of its weight. The diamond broke into pieces in transportation s.t. ratio of wts is 2:3:5. A loss of 840 lakhs was caused. Original price = ?

→ price $\propto w^3$

$p = k w^3$

new weights of pieces : $\frac{2}{10} w, \frac{3}{10} w, \frac{5}{10} w$.

new prices : $k \cdot \frac{8}{1000} w^3, k \cdot \frac{27}{1000} w^3, k \cdot \frac{125}{1000} w^3$

$\frac{8}{1000} kw^3, \frac{27}{1000} kw^3, \frac{125}{1000} kw^3$

total $= \frac{160}{1000} kw^3$

$kw^3 - \frac{160}{1000} kw^3 = 840$ lakhs

$\left(\frac{840}{1000}\right) kw^3 = 840$ lakhs

$\Rightarrow kw^3 = 1000$ lakhs (original price)

1000,00,000

$\Rightarrow$ 10,00,00,000

10 crores

Q. If a glass ball of 55 kg is broken into 3 integral wts, which cannot be ratio of wts ?

a) 1:2:2 b) 2:3:5 c) 3:7:1 d) 1:4:6

→ a) $5x = 55$ ✓

b) $10x = 55$ $x = 5.5$ ✗ b can't give integral weights.

c) $11x = 55$ ✓

d) $11x = 55$ ✓

Q. Ratio of A's salary in Jan to Feb = 1.5 : 1.333... Feb to march = 2 : 2.666...

A got ₹90 more for march than for Jan

and a bonus constituting 40% of salary for 3 months. Find the bonus.

→ Jan : Feb = $\frac{3}{2} : \frac{4}{3} = 9 : 8$ Feb : march = $2 : \frac{8}{3} = 6 : 8 = 3 : 4$

$\begin{array}{r} 9:8 \\ 3:4 \\ \hline 27:24:32 \end{array}$

$27x + 90 = 32x$

$90 = 5x$

$x = 18$

$x = 2.6666$

$10x = 26.666$

$9x = 24$

$x = \frac{24}{9} = \frac{8}{3}$

Total sal $= (27 + 24 + 32)18 = 83 \cdot 18 = 1494$

Bonus $= 40\%$ of $1494 = 597.6$